

§5.2 控制算法/ Control Algorithm

◆ 控制算法为带输入和输出逻辑模块/ The control algorithm is the blocks of input–output system

◆ 单片机的计算能力决定能够采用的控制算法的复杂程度/ The complexity of algorithm is restricted by the computational capability of the MCU of AECs

◆ 典型的控制算法包括/The typical algorithm blocks include:

- 简单的加减乘除数学逻辑/ Simple math operation such as +–x/,…
- 与、或、非等逻辑/Simple logic operation such as and, or, not ,…
- 信号滤波逻辑/ The filtering algorithm of inputs or outputs
- PID等控制逻辑/ The PID control algorithm
- 查MAP图和插值逻辑/ The interpolation or look up tables
- 限制和饱和逻辑/The non–linear blocks such as saturation, limitation et al.

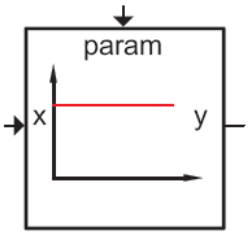
实际的控制算法都要考虑物理意义的极限边界值/ The actual control algorithm should considered the absolute limitation of physical models.

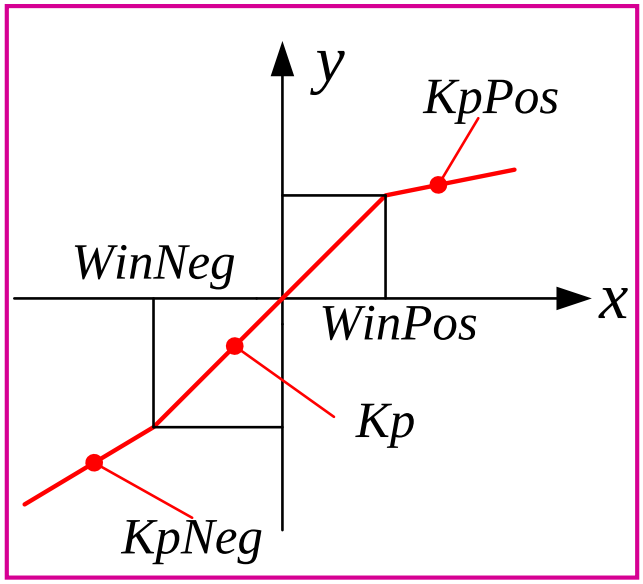


§5.2 控制算法/ Control Algorithm

● 比例控制P/ Proportional element (P)

- ◆ 输入参数：x
- ◆ 输出参数：y
- ◆ 标定参数：Kp，KpPos，KpNeg，WinPos，WinNeg





$$WinNeg \le x_k \le WinPos \qquad y_k = Kp \cdot x_k$$

$$x_k > WinPos, \qquad y_k = Kp \cdot WinPos + KpPos \cdot (x_k - WinPos)$$

$$x_k < WinNeg, \qquad y_k = Kp \cdot WinNeg + KpNeg \cdot (x_k - WinNeg)$$



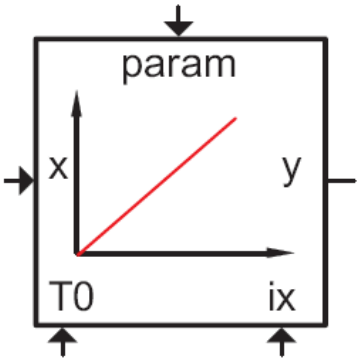
§5.2 控制算法/ Control Algorithm

●积分控制I/ Integral element (I)

◆输入参数x，输出参数y，T0为采样时间，ix为初值；标定参数为Ki，KiPos，KiNeg，WinPos，WinNeg

$$\frac{Ki}{s}$$

$$\frac{Q\cdot z^{-1}}{1-z^{-1}},\quad \Leftarrow Q=Ki\cdot T0$$

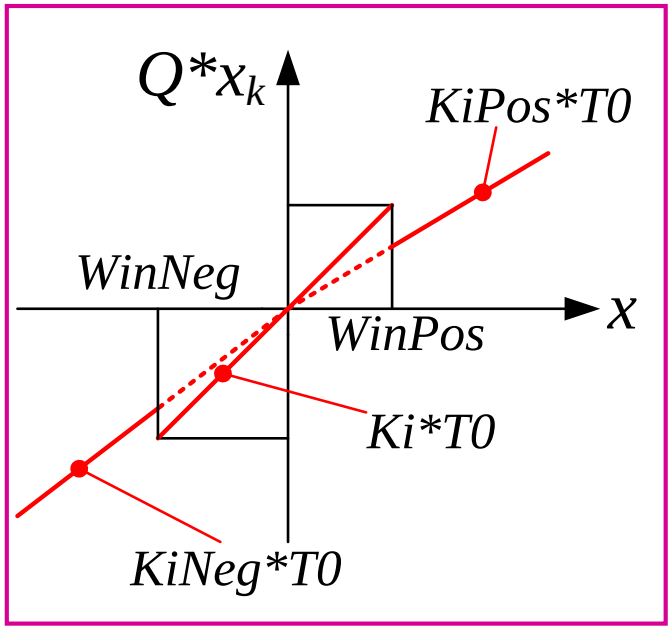


$$y_k = y_{k-1} + Q\cdot x_k$$

$$WinNeg \le x_k \le WinPos,\quad Q=Ki\cdot T0$$

$$x_k > WinPos,\quad Q=KiPos\cdot T0$$

$$x_k < WinNeg,\quad Q=KiNeg\cdot T0$$





§5.2 控制算法/ Control Algorithm

●比例积分控制PI / Porportional and integral element (PI)

◆输入参数x，输出参数y，输出参数yp，输出参数yi，T0为采样事件，ix为初值；标定参数为Ki, KiPos, KiNeg, Kp, KpPos, KpNeg, WinPosI, WinNegI, WinPosP, WinNegP, 以及标定参数lim（边界限制值），pv（输出的预置初值）

$$WinNegP \le x_k \le WinPosP \qquad yp_k = Kp \cdot x_k$$

$$x_k > WinPosP, \qquad y_k = Kp \cdot WinPosP + KpPos \cdot (x_k - WinPosP)$$

$$x_k < WinNegP, \qquad y_k = Kp \cdot WinNegP + KpNeg \cdot (x_k - WinNegP)$$

$$yi_k = yi_{k-1} + Q \cdot x_k, \qquad Q = Ki \cdot T0$$

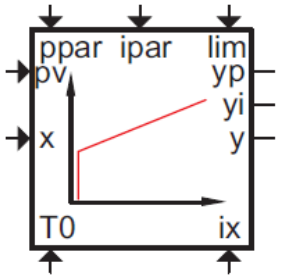
$$WinNegI \le x_k \le WinPosI, \quad Q = Ki \cdot T0$$

$$x_k > WinPosI, \quad Q = KiPos \cdot T0$$

$$x_k < WinNegI, \quad Q = KiNeg \cdot T0$$

$$y_k = yi_k + yp_k$$

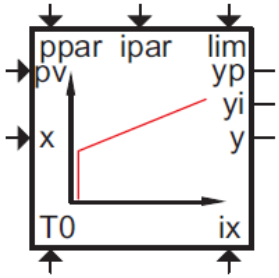
$$yLimNeg \le y_k \le yLimPos$$

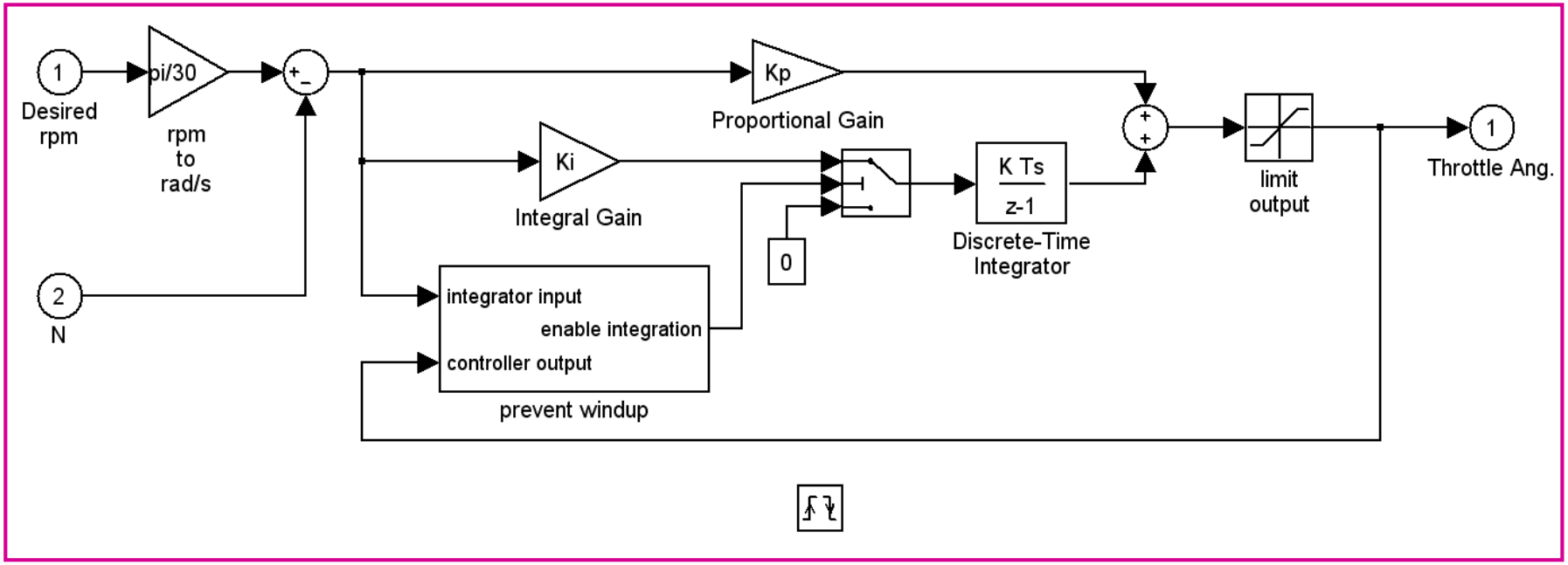




§5.2 控制算法/ Control Algorithm

●比例积分控制PI / Porportional and integral element (PI)







§5.2 控制算法/ Control Algorithm

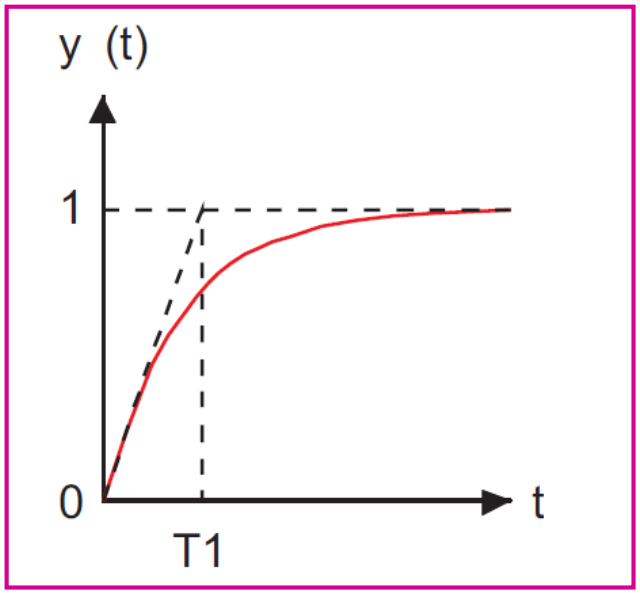
●带1阶延迟的比例环节PT1/ Proportional element with 1st order time delay (PT1)

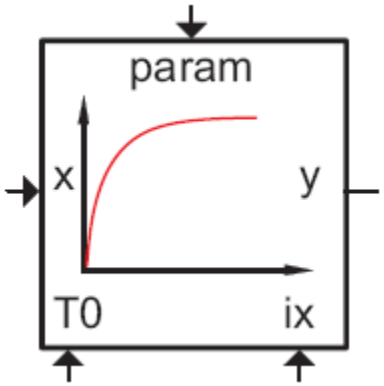
- ◆ 又叫一阶滤波器
- ◆ 输入参数x，输出参数y，T0为采样时间，Ix为初值；标定参数为Q

$$\frac{1}{1+T_1 \cdot s}$$

$$\frac{(1-Q)\cdot z^{-1}}{1-Q\cdot z^{-1}},\quad Q = e^{-\frac{T0}{T1}}$$

$$y_k = Q\cdot y_{k-1} + (1-Q)\cdot x_{k-1}$$







§5.2 控制算法/ Control Algorithm

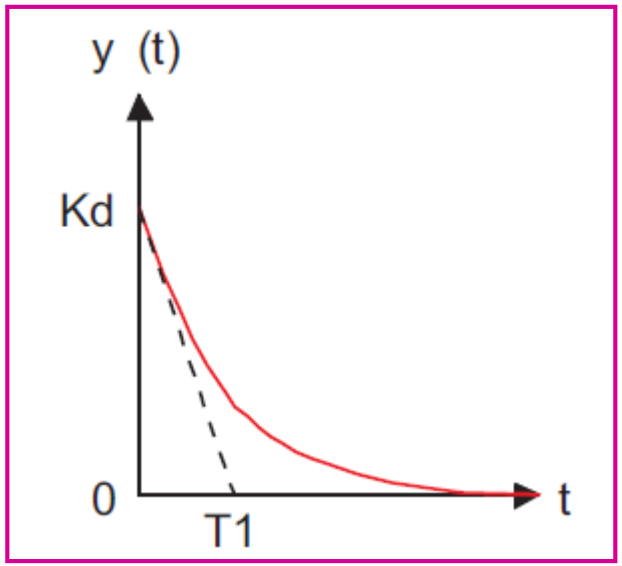
●带1阶延迟的微分环节DT1/ Differential element with 1st order time delay (DT1)

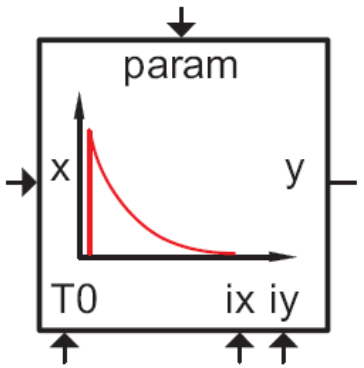
- ◆ 输入参数：x，输出参数：y，T0为采样频率
- ◆ 初始值：ix，iy
- ◆ 标定参数：Kd，Q

$$\frac{Kd \cdot T_1 \cdot s}{1 + T_1 \cdot s}$$

$$\frac{Kd(1 - z^{-1})}{1 - Q \cdot z^{-1}}, \quad Q = e^{-\frac{T0}{T1}}$$

$$y_k = Q \cdot y_{k-1} + Kd \cdot (x_k - x_{k-1})$$





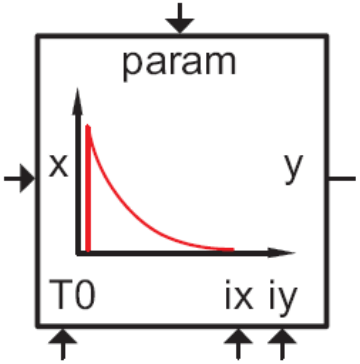


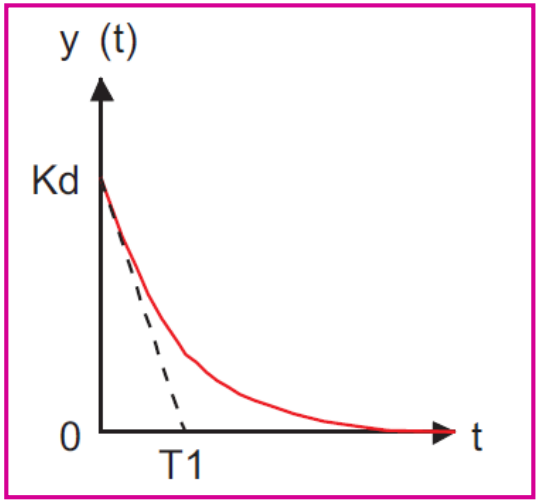
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●带1阶延迟的微分环节（带窗口修正）DT1Win / Differential element with 1st order time delay and slope dependent window (DT1Win)

- ◆ 输入参数：x，输出参数：y，T0为采样频率
- ◆ 初始值：ix，iy
- ◆ 标定参数：Kd，Q，

WinPos, WinNeg, KdPos, KdNeg





$$\frac{Kd \cdot T_1 \cdot s}{1+T_1 \cdot s} \qquad \frac{Kd(1-z^{-1})}{1-Q \cdot z^{-1}},\quad Q = e^{-\frac{T0}{T1}}$$

$$WinNeg \le x_k - x_{k-1} \le WinPos \qquad y_k = Q \cdot y_{k-1} + Kd \cdot (x_k - x_{k-1})$$

$$WinPos < x_k - x_{k-1} \qquad y_k = Q \cdot y_{k-1} + Kd \cdot WinPos + KdPos \cdot (x_k - x_{k-1} - WinPos)$$

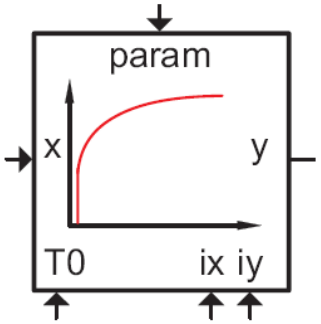
$$WinNeg > x_k - x_{k-1} \qquad y_k = Q \cdot y_{k-1} + Kd \cdot WinNeg + KdNeg \cdot (x_k - x_{k-1} - WinNeg)$$

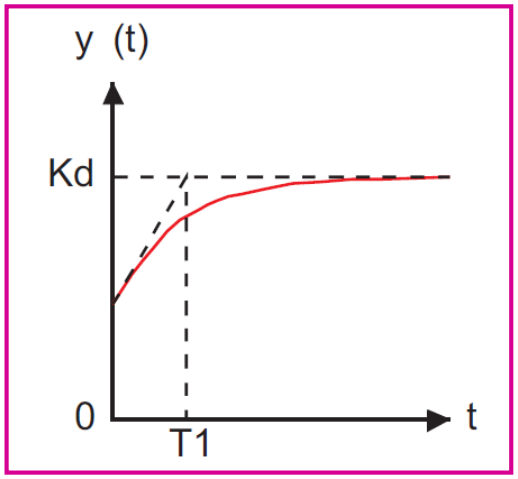


§5.2 控制算法/ Control Algorithm

●带1阶延迟的比例微分环节PDT1 / Proportional and differential element with 1st order time delay (PDT1)

- ◆ 输入参数：x，输出参数：y，T0为采样频率
- ◆ 初始值：ix，iy
- ◆ 标定参数：Kd，Q1，





$$\frac{1+Kd\cdot T_1\cdot s}{1+T_1\cdot s}$$

$$\frac{Kd+Q_2\cdot z^{-1}}{1-Q_1\cdot z^{-1}},\quad Q_1=e^{-\frac{T0}{T1}}\quad Q_2=1-Kd-Q_1$$

$$y_k=Q_1\cdot y_{k-1}+Kd\cdot x_k+Q_2x_{k-1}$$

$$y_k=Q_1\cdot y_{k-1}+Kd\cdot x_k+(1-Kd-Q_1)x_{k-1}$$

$$y_k=Q_1\cdot y_{k-1}+Kd(x_k-x_{k-1})+(1-Q_1)x_{k-1}$$

PT1: $y_k=Q_1\cdot y_{k-1}+(1-Q_1)\cdot x_{k-1}$

DT1: $y_k=Q_1\cdot y_{k-1}+Kd\cdot(x_k-x_{k-1})$

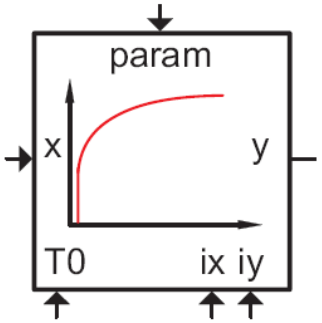


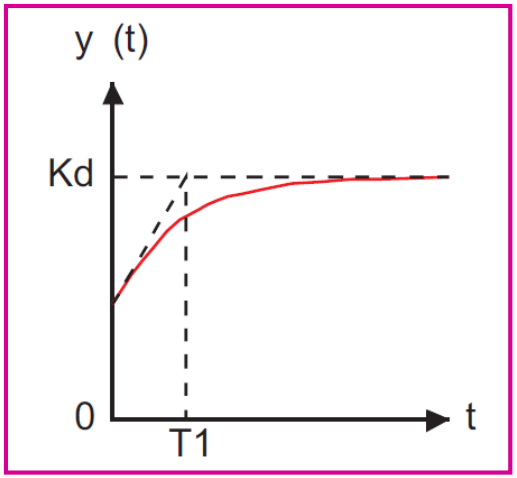
§5.2 控制算法/ Control Algorithm

●带1阶延迟的比例微分环节（带窗口修正）PDT1 Win/ Proportional and differential element with 1st order time delay and slope–dependent window (PDT1Win)

- ◆ 输入参数：x，输出参数：y，T0为采样频率
- ◆ 初始值：ix，iy
- ◆ 标定参数：Kd，Q1，

WinPos, WinNeg, KdPos, KdNeg





$$y_k = Q_1 \cdot y_{k-1} + Kd(x_k - x_{k-1}) + (1-Q_1)x_{k-1}$$

$$WinNeg \le x_k - x_{k-1} \le WinPos$$

$$y_k = Q_1 \cdot y_{k-1} + Kd \cdot (x_k - x_{k-1}) + (1-Q_1)x_{k-1}$$

$$WinPos < x_k - x_{k-1} \qquad y_k = Q_1 \cdot y_{k-1} + Kd \cdot WinPos + KdPos \cdot (x_k - x_{k-1} - WinPos) + (1-Q_1)x_{k-1}$$

$$WinNeg > x_k - x_{k-1} \qquad y_k = Q_1 \cdot y_{k-1} + Kd \cdot WinNeg + KdNeg \cdot (x_k - x_{k-1} - WinNeg) + (1-Q_1)x_{k-1}$$



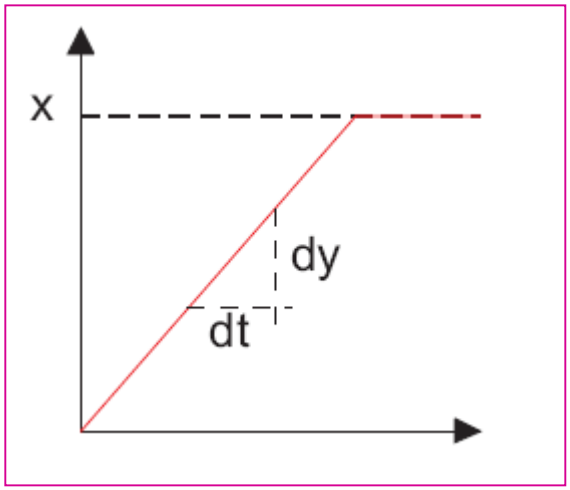
§5.2 控制算法/ Control Algorithm

●斜坡环节/Ramp

$$y_{k-1} < x_k, \quad y_k = y_{k-1} + SlopePos \cdot T0;$$

$$y_{k-1} = x_k, \quad y_k = y_{k-1};$$

$$y_{k-1} > x_k, \quad y_k = y_{k-1} - SlopeNeg \cdot T0;$$



$$y_{k-1} < x_k, \quad y_k = y_{k-1} + StepUp;$$

$$y_{k-1} = x_k, \quad y_k = y_{k-1};$$

$$y_{k-1} > x_k, \quad y_k = y_{k-1} - StepDown;$$

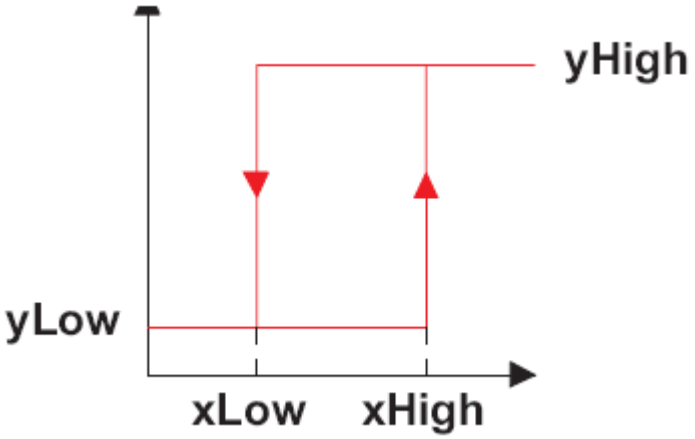


●滞 回 环 节/Hysteretic

$$x_k = xLow, \quad y_k = yLow$$

$$xLow < x < xHigh, \quad y_k = y_{k-1}$$

$$x > xHigh, \quad y_k = yHigh$$

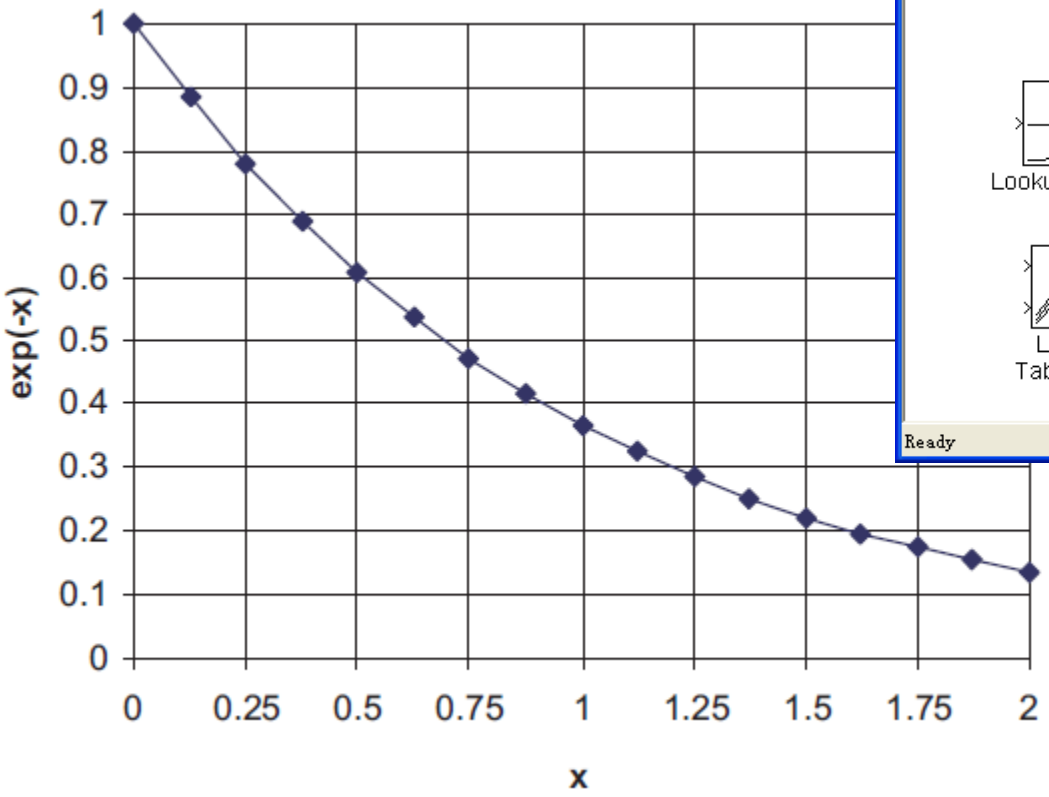


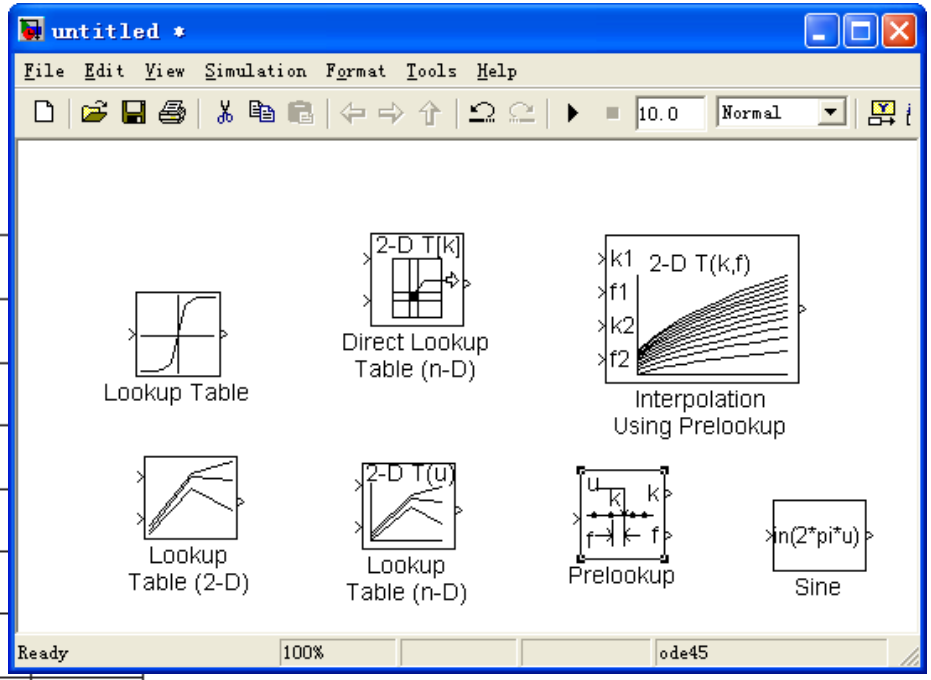


§5.2 控制算法/ Control Algorithm

●插值和查表/Interpolation and Look up table

$$y = e^{-x}$$



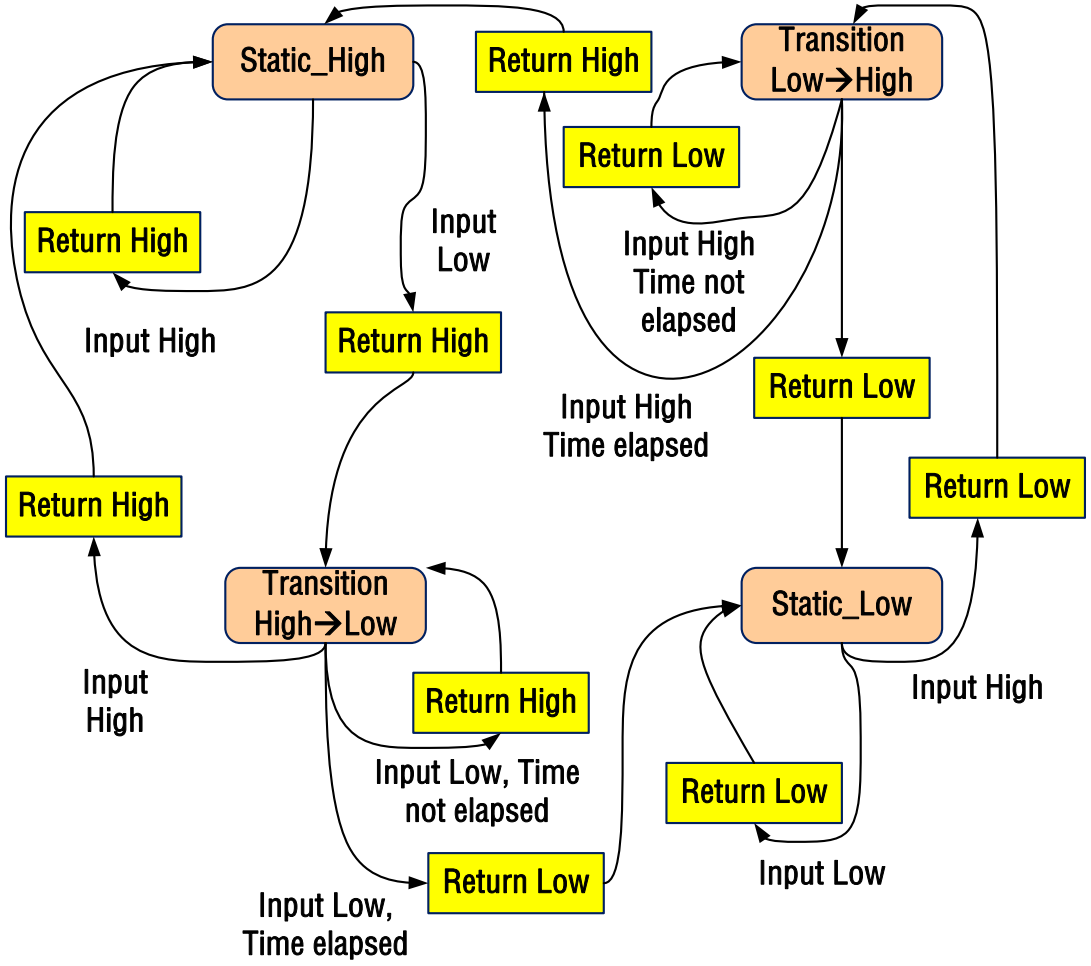




§5.2 控制算法/ Control Algorithm

●反跳/Debounce

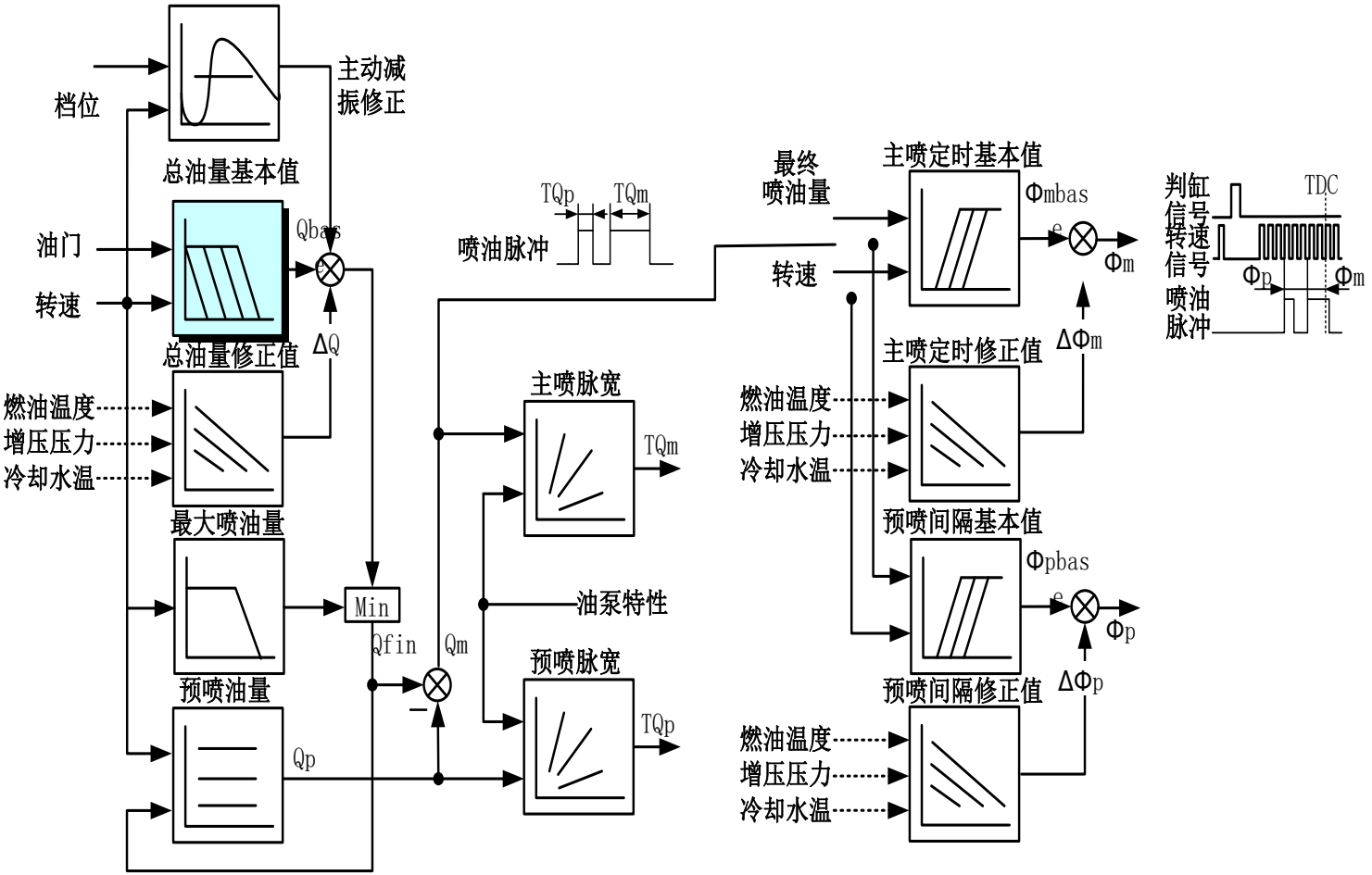
◆故障诊断中经常用的一种算法





§5.2 控制算法/ Control Algorithm

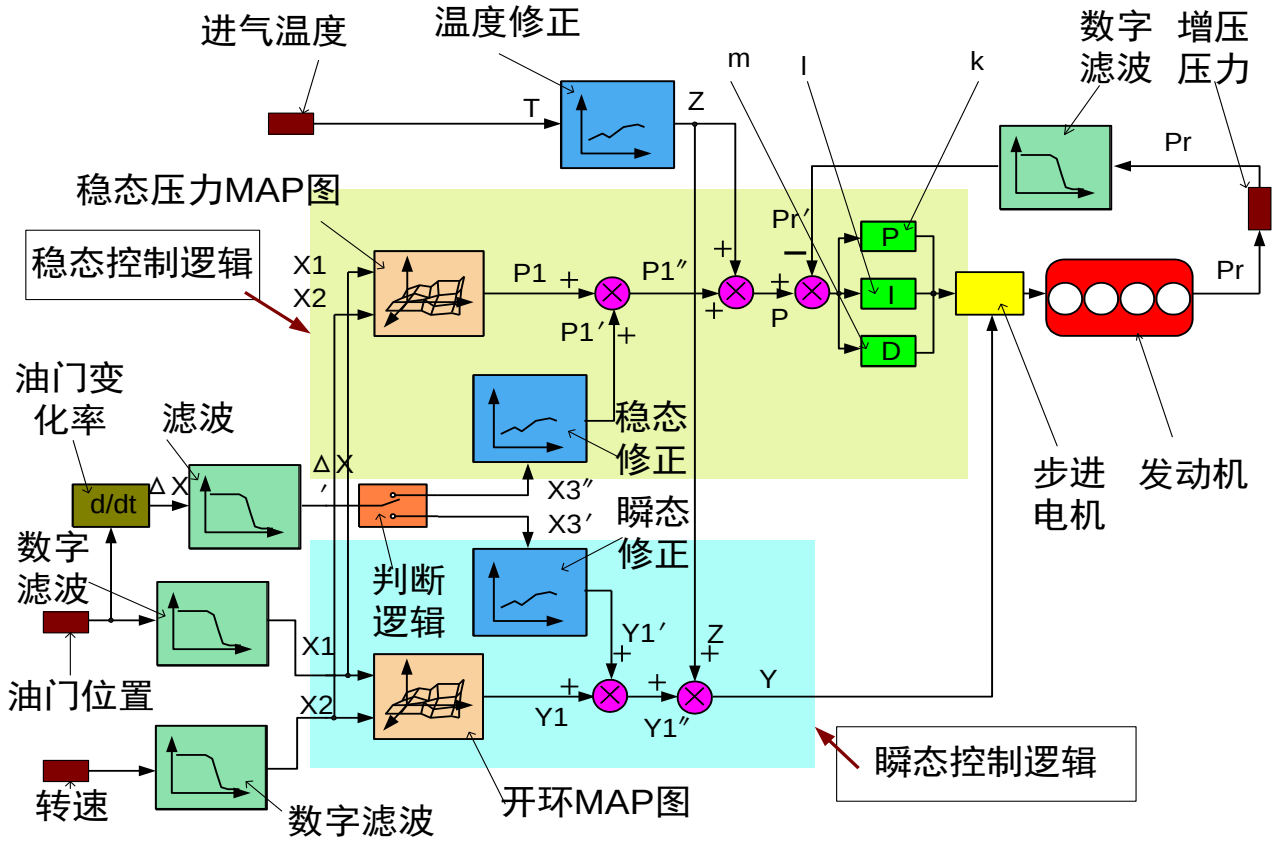
柴油机的燃油管理算法





§5.2 控制算法/ Control Algorithm

◆VNT控制算法逻辑





§5.2 控制算法/ Control Algorithm

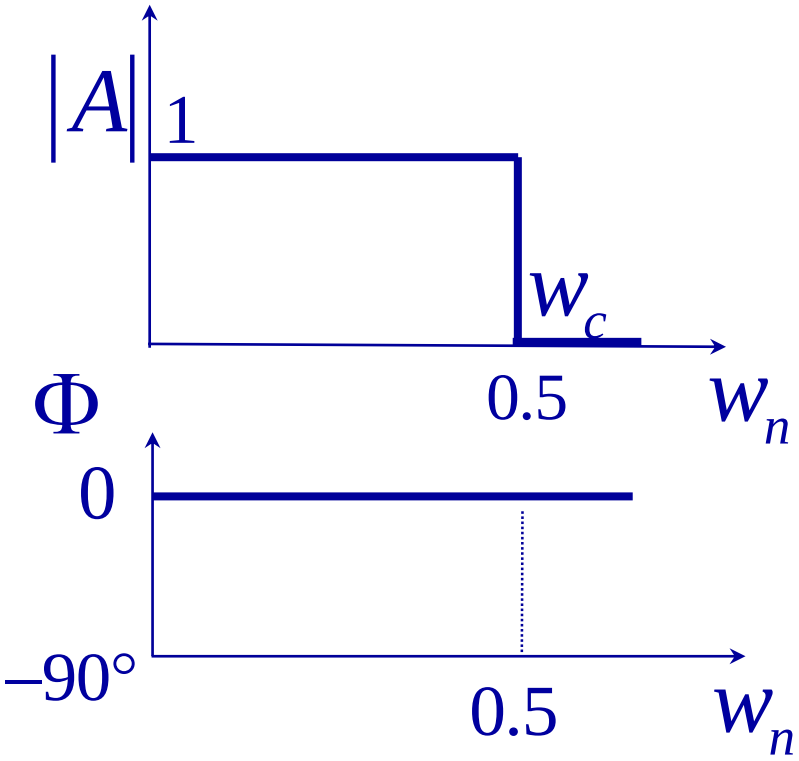
- **理想滤波器：滤波器的传递函数的频率特性**

实际滤波器：

◎存在过渡段；

◎存在相位的滞后；

◎存在波纹；





§5.2 控制算法/ Control Algorithm

- 滤波器的特性表达：看成一个控制系统。
  - 频率特性（BODE图）：幅频特性和相频特性
  - 单位脉冲响应
  - 阶跃响应
- 滤波器的表达方式：
  - 传递函数；
  - 状态空间描述；
  - 零点－极点描述；

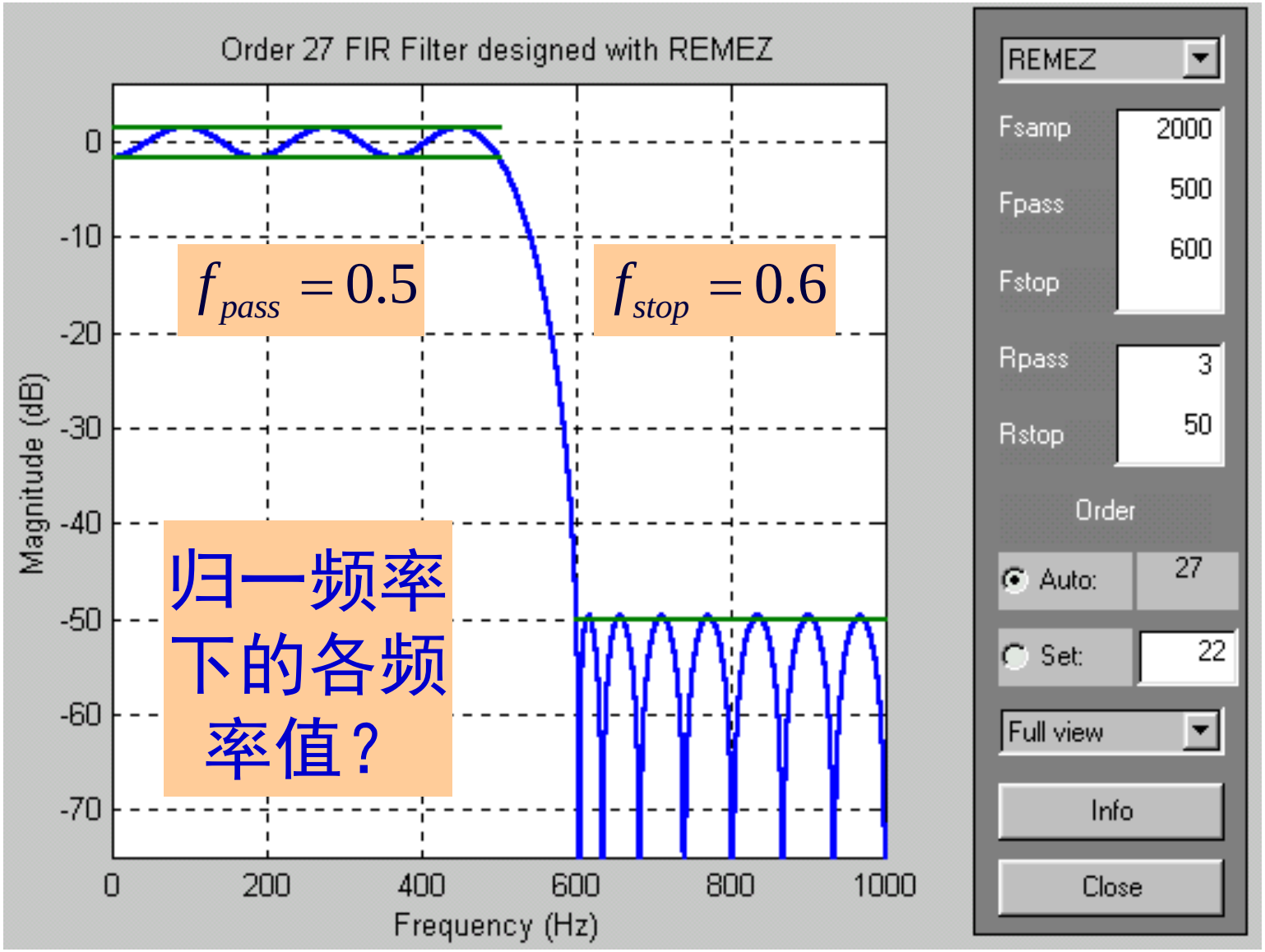




§5.2 控制算法/ Control Algorithm

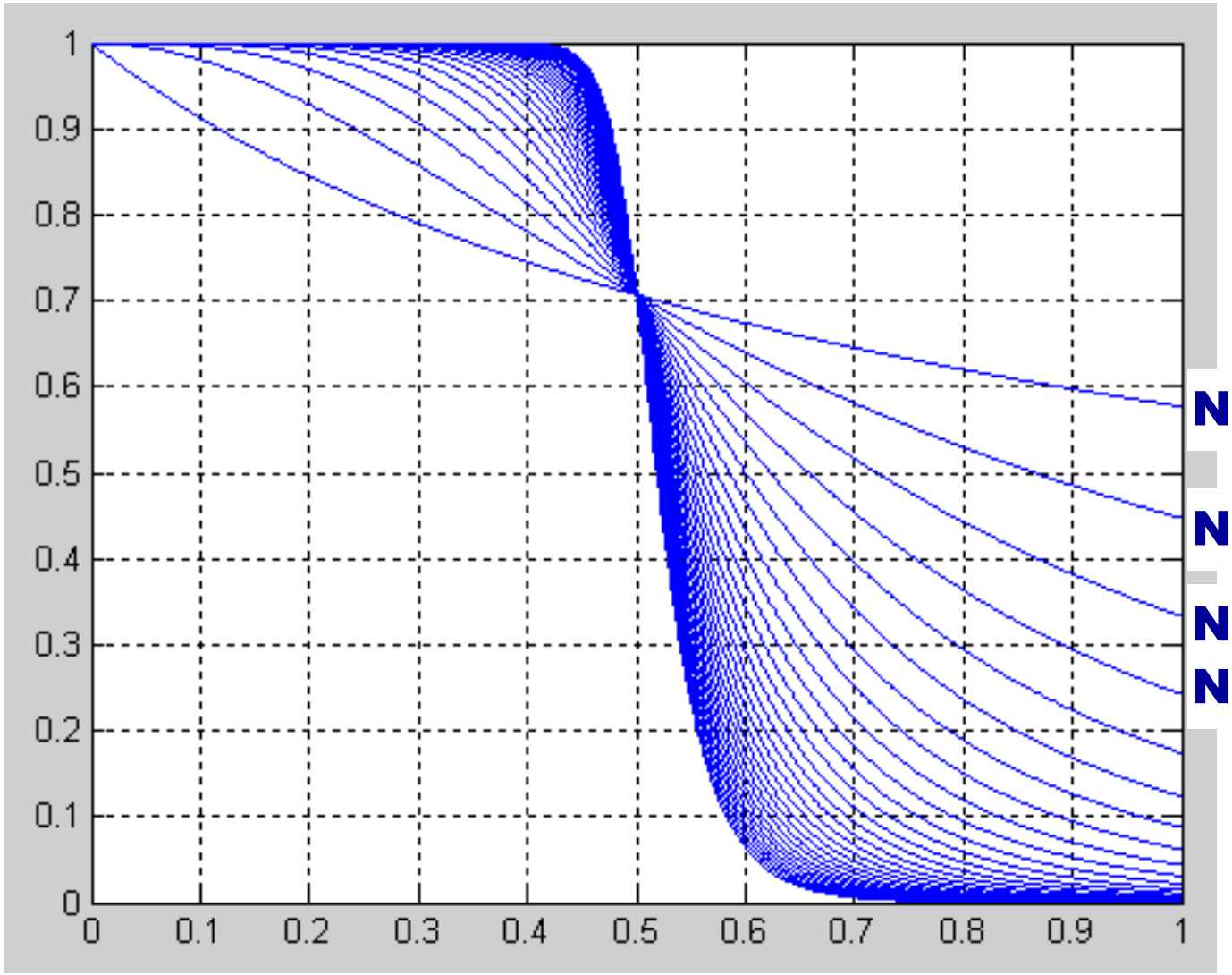
- 过渡段的形状：滤波器的阶次 *N*
- 滤波器的截止频率：Wc
- 通带和止带的宽度
- 通带和止带的波纹的幅度
- 滤波器的分类：
  - FIR：有限冲击响应 Finite Impulse Response
  - IIR：无限冲击响应 Infinite Impulse Response

§5.2 控制算法/ Control Algorithm





§5.2 控制算法/ Control Algorithm



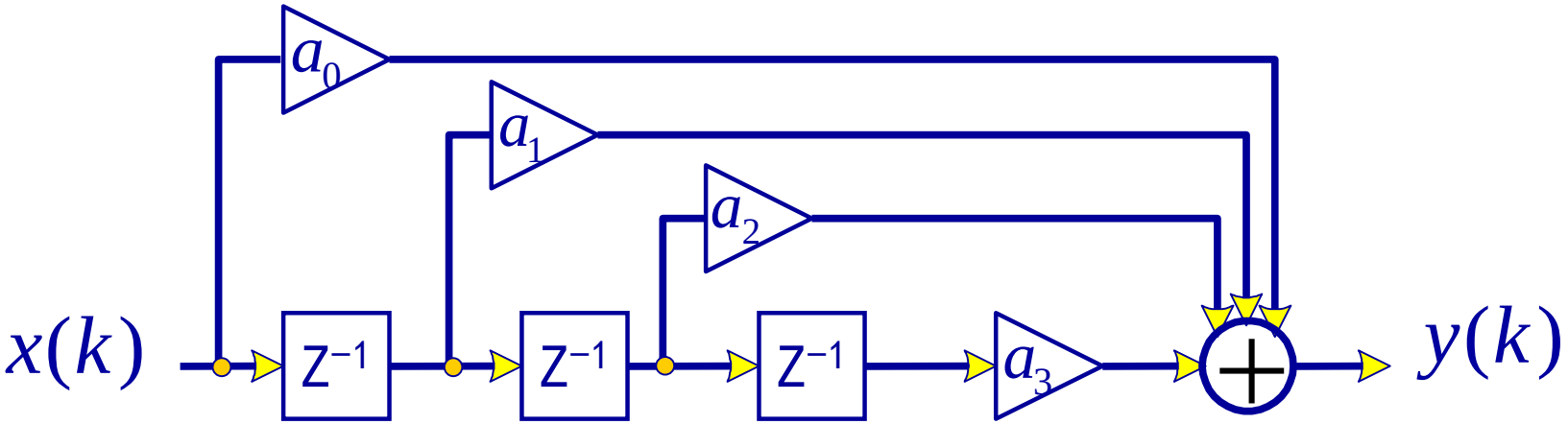
BUTTERW ORTH滤波器的过渡带和阶次**N**的关系



§5.2 控制算法/ Control Algorithm

❑FIR滤波器的含义（有限冲击响应）

$$y(k)=a_0x(k)+a_1x(k-1)+a_2x(k-2)+a_3x(k-3)$$



$$y(z)=a_0x(z)+a_1x(z)z^{-1}+a_2x(z)z^{-2}+a_3x(z)z^{-3}$$

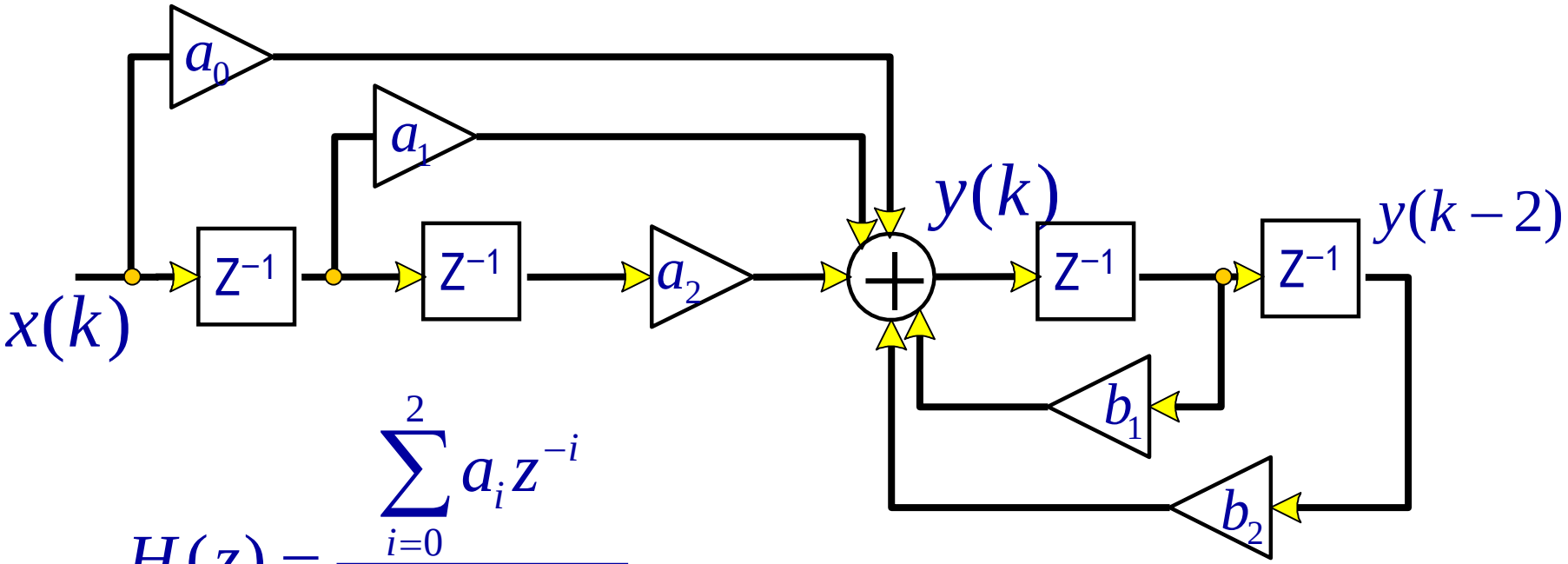
$$H(z)=a_0+a_1z^{-1}+a_2z^{-2}+a_3z^{-3}=\sum_{i=0}^{3}a_iz^{-i}$$



§5.2 控制算法/ Control Algorithm

❑IIR滤波器的含义（无限冲击响应）

$$y(k)=a_0x(k)+a_1x(k-1)+a_2x(k-2)+b_1y(k-1)+b_2y(k-2)$$



$$H(z)=\frac{\sum_{i=0}^{2}a_iz^{-i}}{1-\sum_{i=1}^{2}b_iz^{-i}}$$

$$H(z)=\sum_{i=0}^{3}a_iz^{-i}$$



§5.2 控制算法/ Control Algorithm

- MATLAB工具的介绍
- Fftdemo:快速Fourier变换的例子
- Filtdemo:滤波器的演示部分
- Sptool:滤波器设计/信号分析的工具箱
- DSP Blockset:数字信号处理工具箱

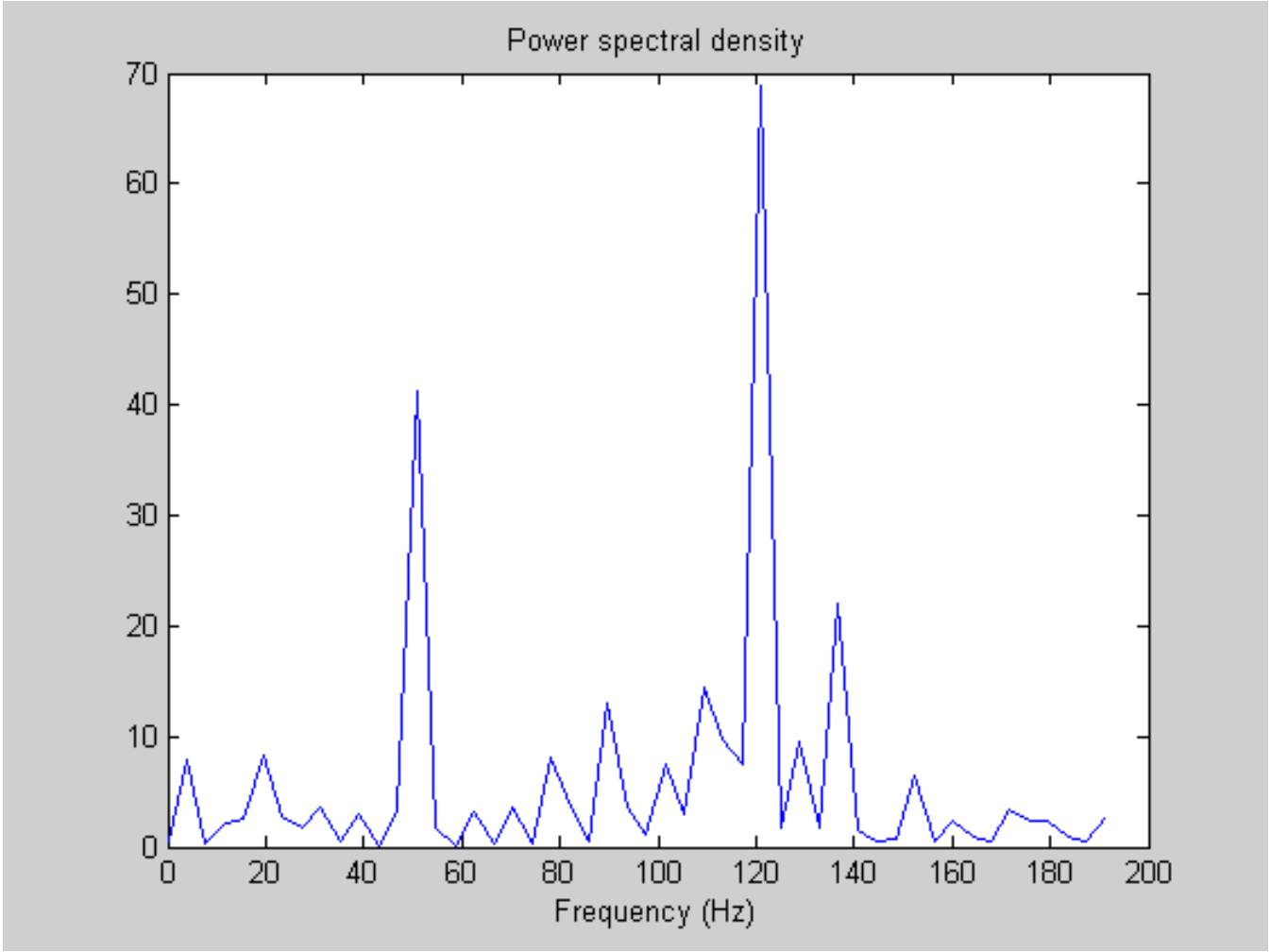


（1）Fftdemo

❑直接在MATLAB COMMAND WINDOW下运行：fftdemo;

问题：如何看到fftdemo源代码？

答案：type fftdemo.m

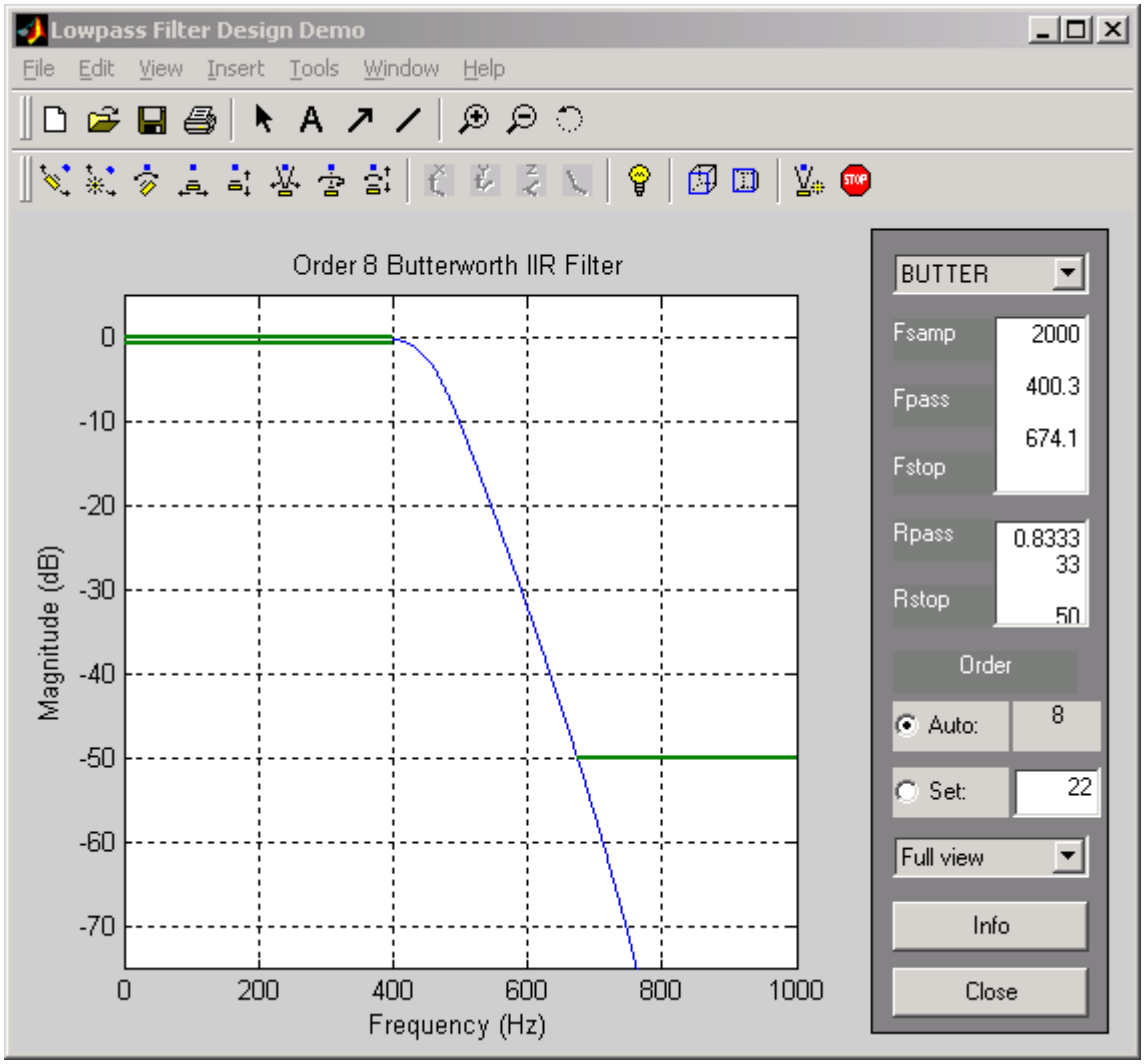




（2）Filtdemo

❑直接在MATLAB COMMAND WINDOW下运行：filtdemo;

利用此工具可以方便演示滤波器的设计过程





（3）SPTOOL

- SPTOOL是MATLAB工具箱中专门用于信号处理的工具箱，具有较好的功能，推荐大家使用。

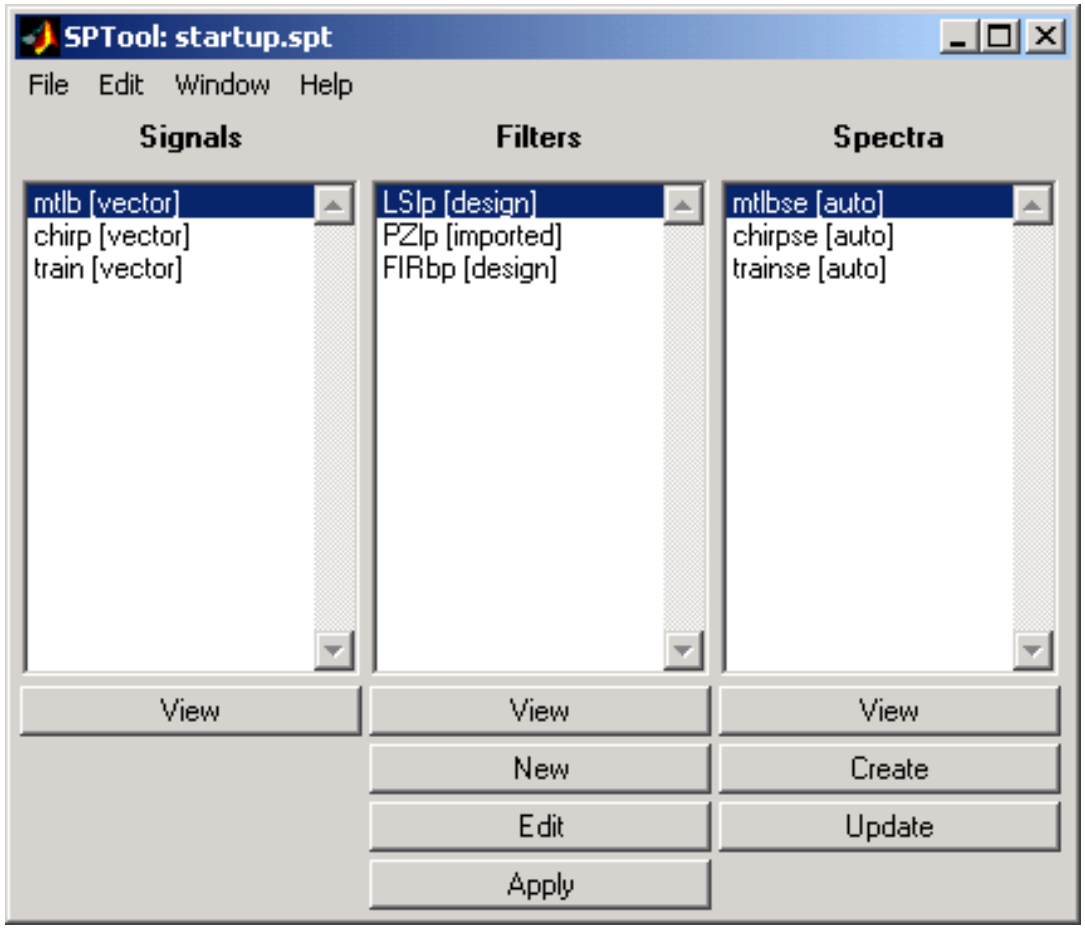
- 直接在MATLAB COMMAND WINDOW下运行：sptool;

- 进入SPTOOL工具箱。



（3）SPTOOL

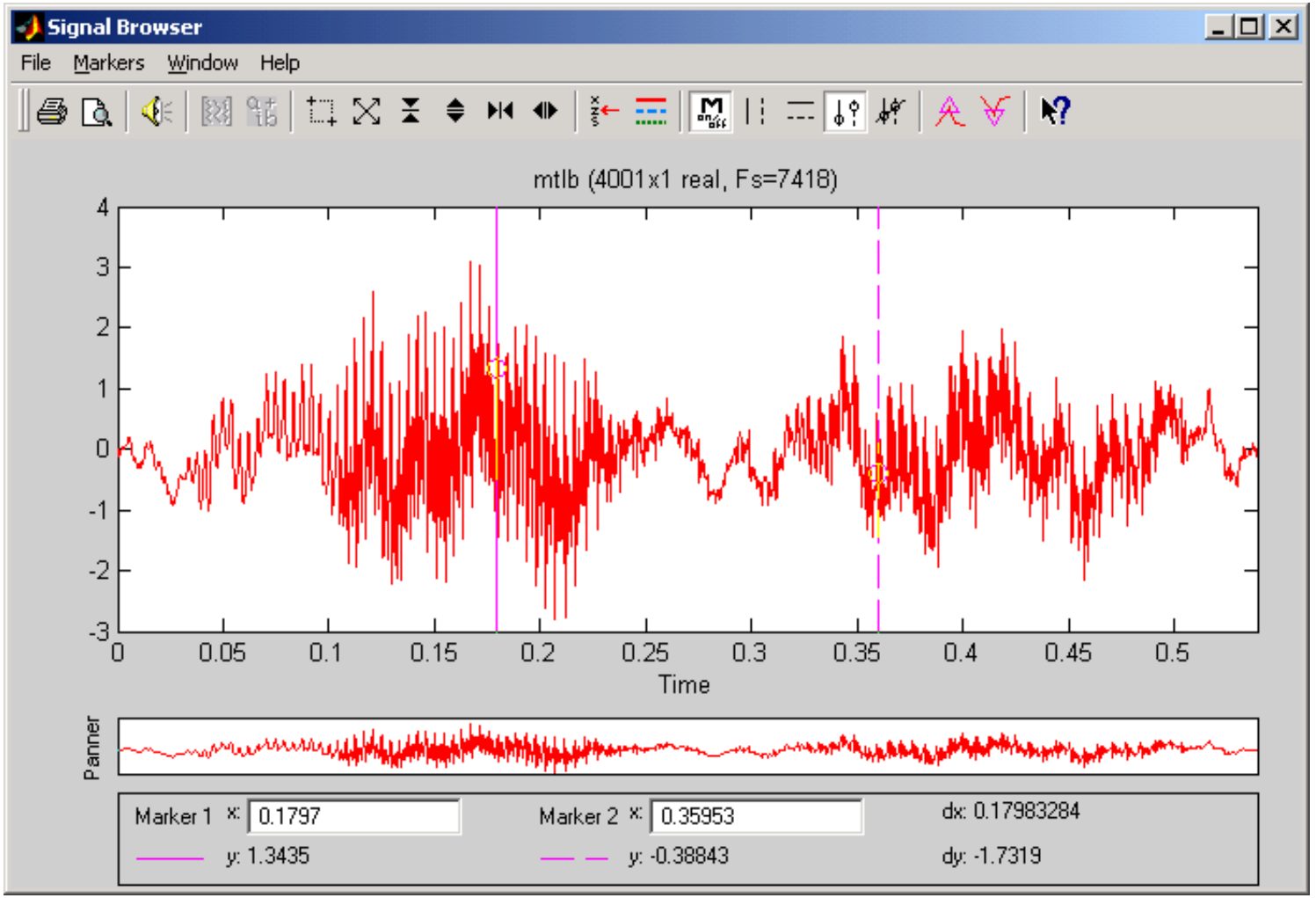
❑SPTOOL包括三个部分：信号时域分析（左）、滤波器设计（中）、信号频域分析（右）





（3）SPTOOL

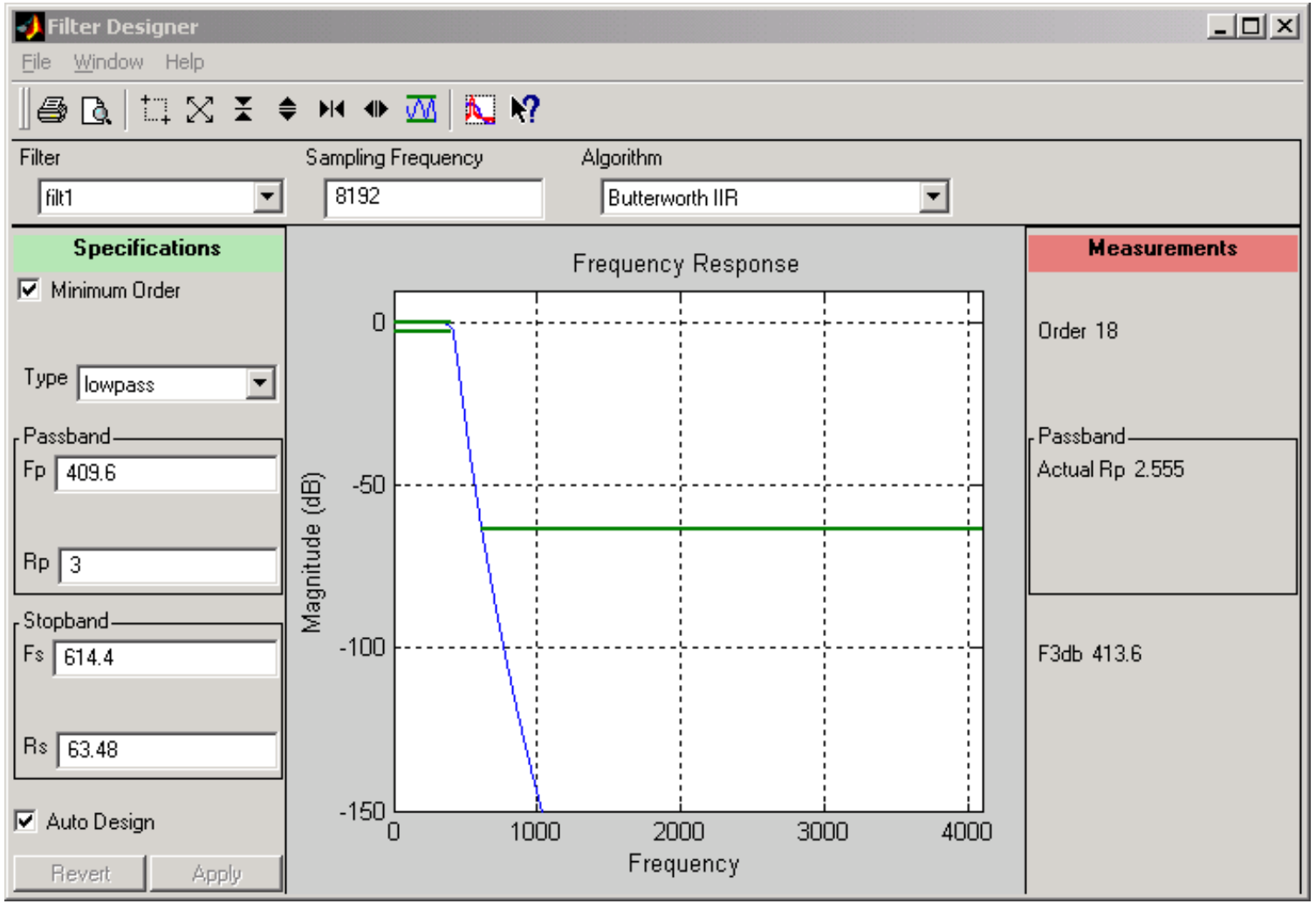
❑信号时域分析能够对信号进行统计和观察





（3）SPTOOL

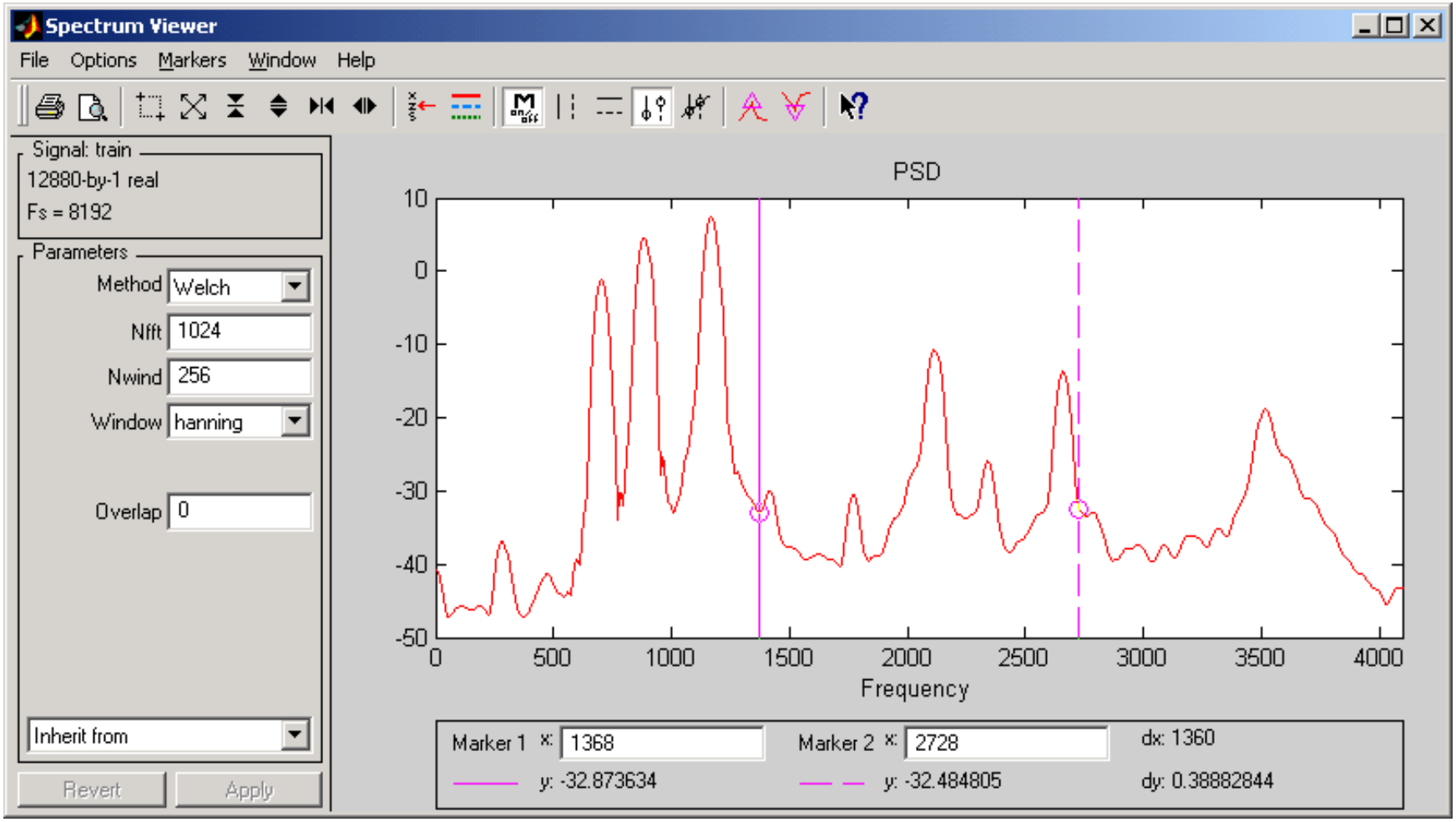
❑滤波器设计部分能够观察和设计需要的滤波器





（3）SPTOOL

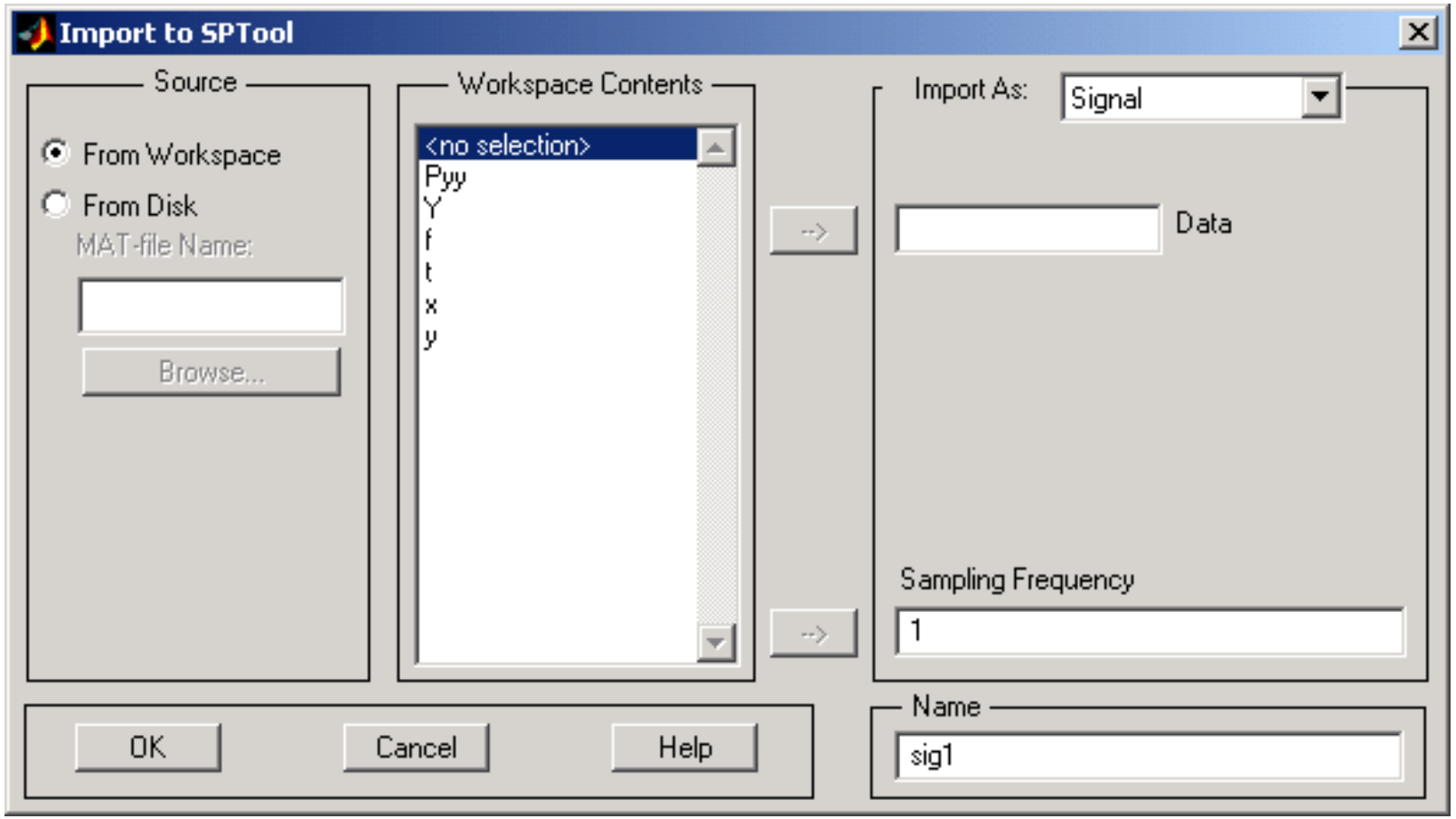
❑信号频域分析能够观察和分析信号在频域的特征





（3）SPTOOL

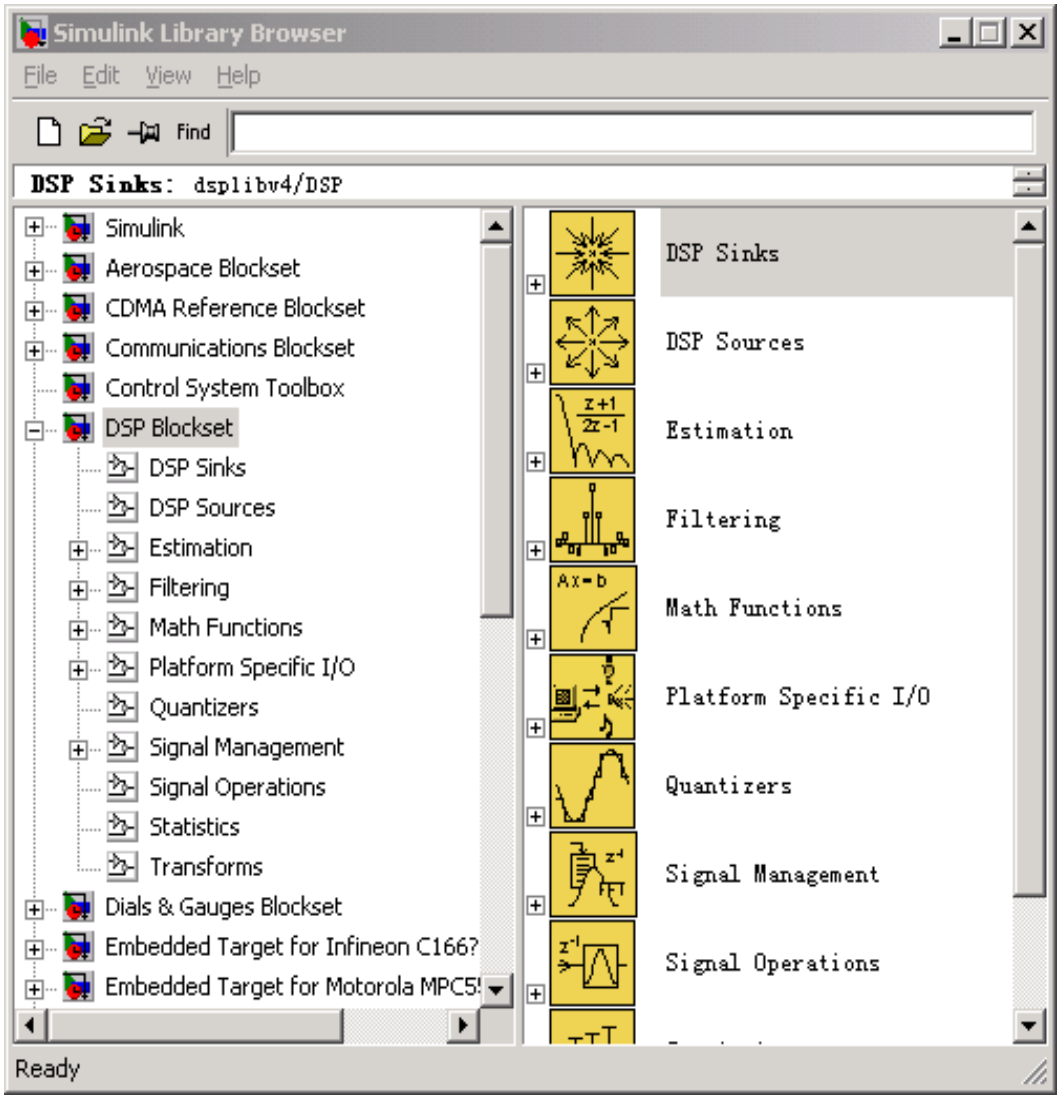
❑信号、滤波器能够方便地从MATLAB的workspace调入和调出到SPTOOL，为信号处理提供便利的条件；





（4）DSP Blockset

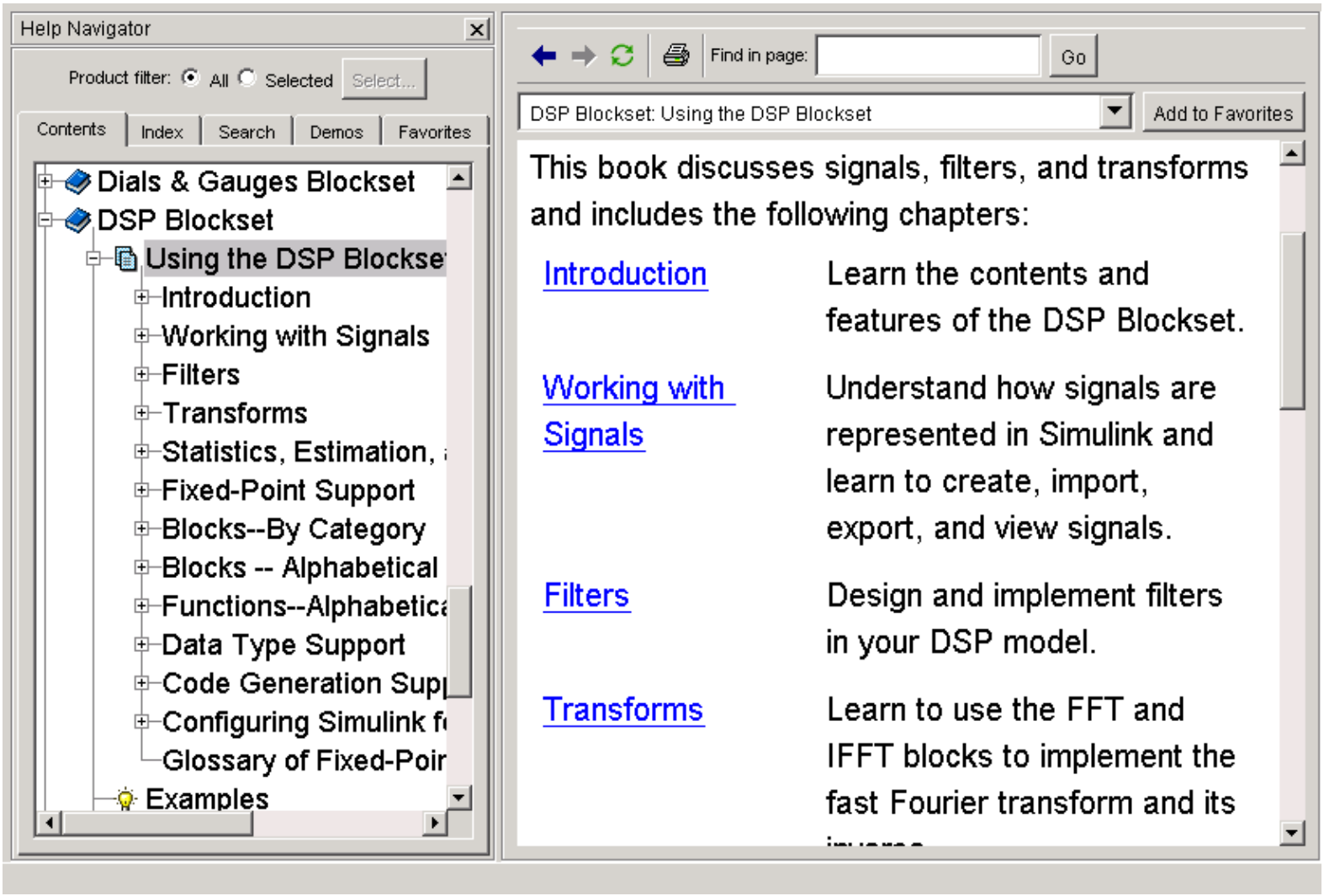
- DSP Blockset是SIMULINK中信号处理工具箱之一，能够的非实时信号处理和实时在线信号处理的功能；所开发的信号处理算法可以直接下载到单片机中





（4） DSP Blockset

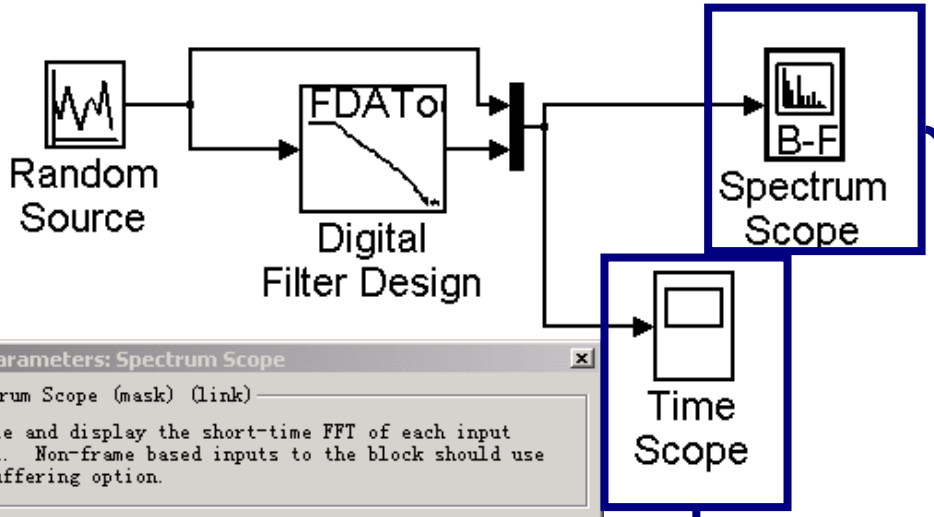
❑请自学DSP Blockset的help文档！

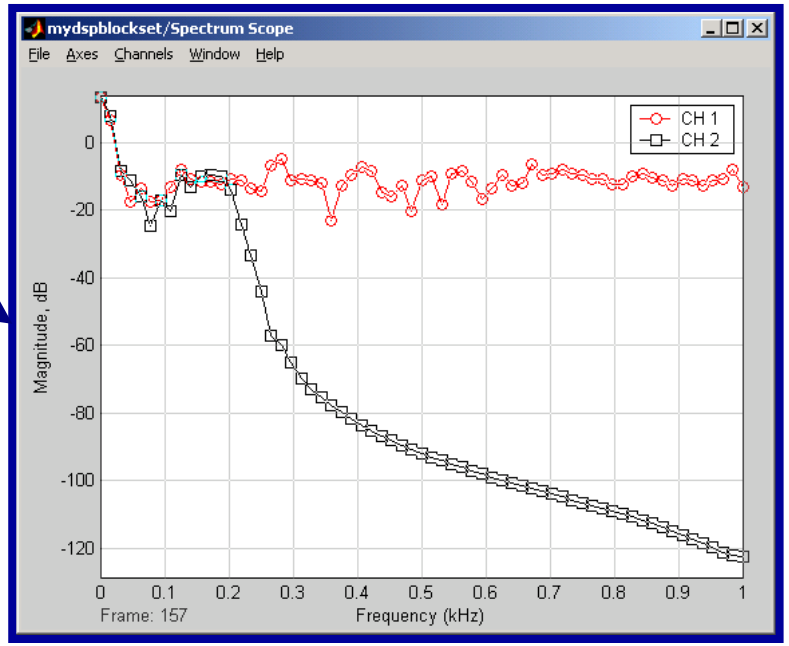


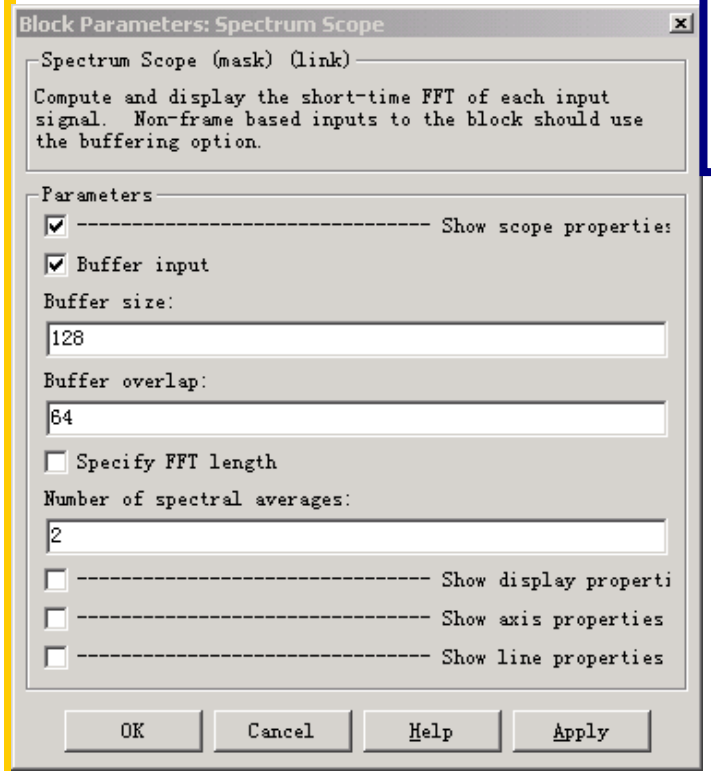


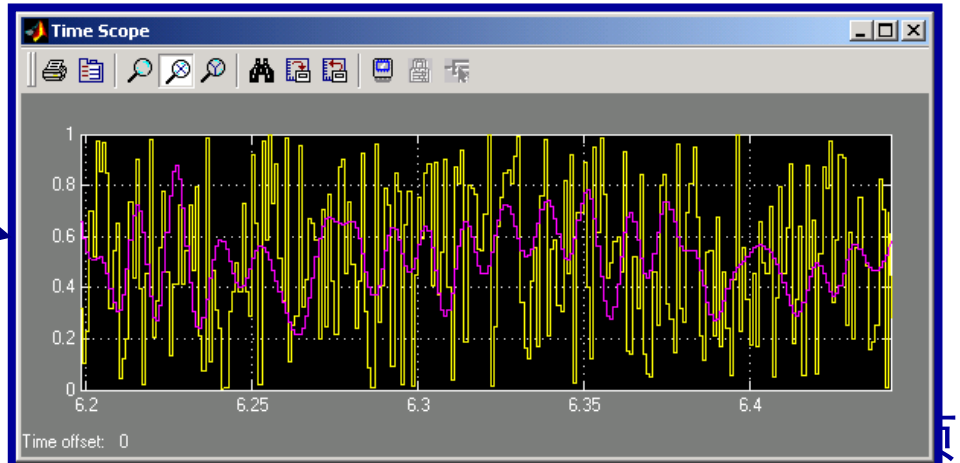
（4） DSP Blockset

❑DSP Blockset举例：频域示波器











❑一个简单的滤波器：

$$y(k)=a_0x(k)+a_1x(k-1)+b_1y(k-1)+b_2y(k-2)$$

$$y(k)=0.25x(k)+0.25x(k-1)+0.25y(k-1)+0.25y(k-2)$$

$$H(z)=\frac{0.25+0.25z^{-1}}{1-0.25z^{-1}-0.25z^{-2}}=\frac{0.25z^2+0.25z}{z^2-0.25z-0.25}$$